

WANIYA FAROOQ

AI & Data Engineering Student

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PROFESSIONAL SUMMARY

AI undergraduate at COMSATS University Lahore (CGPA: 3.73/4.00) with hands-on experience building AI-powered applications and data pipelines. Proficient in SQL, ETL processes, database design, and end-to-end ML/LLM integration. Experienced with cloud platforms (Azure fundamentals) and data warehousing concepts. Certified through Harvard CS50 and Coursera IBM. Strong foundation in DSA with extensive LeetCode practice.

EDUCATION

BS Artificial Intelligence | COMSATS University Islamabad, Lahore Campus

2024 – Present

CGPA: 3.73 / 4.00

Relevant Coursework: OOP, Data Structures & Algorithms, Database Systems, Machine Learning, Linear Algebra, Statistics

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, HTML/CSS

Data Engineering & ETL: ETL Pipeline Design, Data Wrangling, Data Warehousing Concepts, SQL Querying, Null-value Imputation, Data Validation, CSV/JSON Processing

Databases: SQLite, MySQL, SQLAlchemy ORM, Relational Schema Design (9-table normalized DB), CRUD Operations

Cloud & Platforms: Microsoft Azure (Fundamentals), Git, GitHub, VS Code, Jupyter Notebook

ML / Data Libraries: scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Web Frameworks: Flask, Django, SQLAlchemy, WTForms, Jinja2, Streamlit

PROJECTS

Netflix Dataset —ETL & Exploratory Data Analysis

Python | Pandas | NumPy | Seaborn | Matplotlib | Jupyter Notebook

- Designed and executed an end-to-end ETL pipeline on 100+ Netflix records: extracted raw data, performed data wrangling and null-value imputation across 5 critical features, and loaded clean data for analysis.
- Engineered 6 statistical visualizations (histogram, pie, scatter, bar, multi-subplot) uncovering content distribution patterns, rating segmentation, release-year trends, and geo-based content density across 10 top-producing countries.
- Applied data warehouse-style schema thinking to organize and query tabular data, enabling structured reporting on content metadata.

AI Smart Attendance System — Face Recognition-Based Automation

Python | OpenCV | LBPH | Streamlit | Pandas | Matplotlib

- Built end-to-end face recognition pipeline using OpenCV Haar Cascade + LBPH algorithm for real-time student detection and automatic attendance marking with <70ms processing latency.
- Architected modular system with 5 core classes (StudentManager, AttendanceManager, FileHandler, FaceRecognizer) demonstrating clean OOP design, separation of concerns, and production-grade error handling.
- Implemented multi-layer duplicate prevention via session tracking, CSV validation, and cooldown mechanisms — ensuring data integrity across concurrent webcam streams (ETL-style data quality controls).
- Developed interactive Streamlit dashboard with 7 pages, real-time PDF export, Pandas analytics, and Matplotlib visualizations supporting 50+ student datasets with heatmaps and attendance trend analysis.

Hotel Reservation System — Full-Stack Web Application

Python | Flask | SQLAlchemy | SQLite | WTForms | Jinja2

- Built a full-stack application with 2 role-based portals (Admin & Customer) using Flask Blueprint architecture for separation of concerns.
- Designed a 9-table normalized relational database in SQLite with SQLAlchemy ORM handling 100% of CRUD operations — demonstrating strong database schema design and SQL proficiency.
- Applied ETL principles to automate staff onboarding: extracted form inputs, transformed and validated data, loaded linked Employee + Admin records — cutting admin steps by ~60%.
- Secured both user roles with Werkzeug password hashing and CSRF protection across all forms.

CERTIFICATIONS & COURSES

- CS50's Introduction to AI with Python — Harvard University (edX)
- Agent Skills with Anthropic — DeepLearning.AI
- ChatGPT Prompt Engineering for Developers — DeepLearning.AI
- Data Analysis with Python — Coursera (IBM)